After canceling the terms proportional to $x^2 e^{-\frac{\omega}{2\hbar}x^2}$, we discover the remarkable fact that solving the Schrödinger equation just reduces to solving

$$\frac{\hbar}{2}\omega e^{-\frac{\omega}{2\hbar}x^2} = Ee^{-\frac{\omega}{2\hbar}x^2}.$$

As you can see, the only way we can solve this equation is to set the energy E equal to $\frac{\omega\hbar}{2}$. In other words, we've found not only the wave function but also the value of the ground-state energy. Calling the ground-state energy E_0, we can write

$$E_0 = \frac{\omega\hbar}{2}. \qquad (10.16)$$

The ground-state wave function, meanwhile, is just the Gaussian function the professor gave us:

$$\psi_0(x) = e^{-\frac{\omega}{2\hbar}x^2}.$$

He's a clever fellow, that professor.

10.6 CREATION AND ANNIHILATION OPERATORS

Over the course of these lectures, we have seen two ways of thinking about quantum mechanics. They go all the way back to Heisenberg and Schrödinger. Heisenberg liked algebra, matrices, and, had he known what to call them, linear operators. Schrödinger, by contrast, thought in terms of wave functions and wave equations, the Schrödinger equation being one famous example. Of course, the two ways of thinking are not contradictory; functions form a vector space and derivatives are operators.

So far, in our study of the harmonic oscillator we have focused on functions and differential equations. But the more powerful tool in many cases—particularly for the harmonic oscillator—is the operator method. It